

Hasin Rehana

PhD Candidate, Computer Science · Expected December 2026

Graduate Research Assistant, Department of Biomedical Sciences

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RESEARCH SUMMARY

PhD candidate in Computer Science (defending December 2026) working in **biomedical NLP and large language models** for information extraction from biomedical literature (e.g. protein–protein interactions, vaccine adjuvants and adjuvant mechanisms, and ontology-grounded extraction). Dissertation develops a **multimodal framework for clinical risk prediction**, fusing medical imaging with clinical variables. Multiple publications including first-author work in *Bioinformatics Advances*, *Frontiers in Cellular and Infection Microbiology* and an oral presentation at *AMIA 2025*; 2nd place in the BioASQ12 nested NER challenge. Fluent in PyTorch, LLM fine-tuning and evaluation, and Slurm-managed HPC; active CITI/HIPAA certifications for clinical research.

EDUCATION

Ph.D. in Computer Science

Expected Dec 2026

University of North Dakota, Grand Forks, ND, USA

GPA: 4.00/4.00

- Research focus: Natural Language Processing, medical imaging, and deep learning for biomedical information extraction and clinical decision support, including large language models for biomedical text mining, inference, and scalable fine-tuning on HPC systems.

M.Sc. in Computer Science & Engineering

2022

University of Dhaka, Bangladesh

GPA: 3.73/4.00

- Thesis: Plant leaf disease detection using deep learning (object detection on agricultural imagery with Faster R-CNN).

B.Sc. in Computer Science & Engineering

2018

Rajshahi University of Engineering and Technology (RUET), Bangladesh

GPA: 3.63/4.00

- Thesis: Spatial–spectral CNN feature extraction for hyperspectral image classification.

RESEARCH INTERESTS

Artificial Intelligence in Biomedicine; Biomedical Natural Language Processing (BioNLP); Large Language Models (LLMs); Clinical Imaging; Multimodal Learning; Biomedical Text and Relation Mining; Machine Learning for Health Informatics; Applied Bioinformatics.

TECHNICAL SKILLS

- **Machine Learning:** Multimodal learning, convolutional neural networks, transformer architectures.
- **Imaging & Vision:** Medical imaging, hyperspectral imaging, object detection, image classification.
- **Natural Language Processing:** Biomedical NLP, named entity recognition, relation extraction, transformer-based models (BERT, Llama, Mistral, Qwen, GPT-family).
- **Frameworks & Libraries:** PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV.
- **Infrastructure & Tools:** CUDA, Linux HPC environments, Slurm, Docker, Git, AWS SageMaker, Weights & Biases.
- **Programming:** Python (expert), C++, Java, SQL.

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant — University of North Dakota (Hur Lab)

Jan 2023 – Present

- Conducted large-scale biomedical NLP research using transformer-based models for named entity recognition and relation extraction from scientific literature.
- Developed LLM-based pipelines for protein–protein interaction and vaccine adjuvant extraction.
- Engineered a multilayer BERT-based architecture for nested entity recognition, achieving **2nd place** in the BioASQ12.
- Optimized LLM fine-tuning and inference using GPU-accelerated workflows on Slurm-managed HPC clusters.

Clinical Research Study Team Member — Sanford Research

Jun 2025 – Present

- Research project focused on developing a GPU-accelerated multimodal framework for surgical risk prediction in TAVR.
- Current work includes data understanding, study design, and planning multimodal fusion of medical imaging and longitudinal clinical variables.

Participant — NIAID–BRC AI Codeathon, Argonne National Laboratory

Nov 2025

- Project: PubMed Miner ([GitHub](#)) — An AI-Powered Sequence Feature Extraction from Literature (JCVI).
- Contribution: Developed a natural language query translation module that converts free-text user input into structured PubMed search queries for downstream processing, significantly improving tool usability for domain scientists.

Lecturer — Daffodil International University, Bangladesh

Jan 2019 – August 2022

- Taught undergraduate computer science courses (e.g. Algorithms, Artificial Intelligence, Numerical Methods, and Computer Graphics).
- Mentored undergraduate thesis projects in deep learning and computer vision.

PUBLICATIONS

* indicates equal contribution

1. **H. Rehana***, N.B. Çam*, M. Basmaci, J. Zheng, C. Jemiyo, Y. He, A. Özgür, J. Hur. (2024). *Evaluating GPT and BERT models for protein–protein interaction identification in biomedical text*. *Bioinformatics Advances*, 4(1), vbae133. **(IF 2.6)**
2. J. Monickaraj*, **H. Rehana***, A. Bernardi, Y. Zheng, T. Lin, L. Liu, A. Madireddi, L. Yeh, J. Zheng, J. Hur, Y. He. (2026). *Vaxjo 2.0: An ontology- and large language model-powered knowledge base of vaccine adjuvants and mechanisms*. *Frontiers in Cellular and Infection Microbiology*, 16, 1763384. **(IF 5.5)**
3. J. Zheng, A.Y. Lin, A. Huffman, ..., **H. Rehana**, ..., C. Tao, J. Hur, B. Smith, Y. He. (2026). *VO: The Vaccine Ontology*. *Scientific Data* (Nature Portfolio). doi:10.1038/s41597-026-07298-w. **(IF 7.2)**
4. **H. Rehana**, J. Zheng, F.-Y. Yeh, B. Bansal, N.B. Çam, C. Jemiyo, B. McGregor, A. Özgür, Y. He, J. Hur. (2025). *Leveraging large language models for cancer vaccine adjuvant name extraction from biomedical literature*. *Proceedings of the American Medical Informatics Association (AMIA) Annual Symposium 2025*. (Oral presentation).
5. C. Jemiyo, B.A. McGregor, **H. Rehana**, J. Hur. (2025). *Adverse childhood experiences and chronic health outcomes: evidence from 33 US states in the Behavioral Risk Factor Surveillance System, 2019–2023*. *BMC Public Health*, 25(1), 1650. **(IF 4.4)**
6. **H. Rehana***, B. Bansal*, N.B. Çam, J. Zheng, Y. He, A. Özgür, J. Hur. (2024). *Nested named entity recognition using multilayer BERT-based model*. *CLEF Working Notes*.
7. **H. Rehana**, M.R. Ahmed, R. Chakma, S. Asaduzzaman, M. Raihan. (2021). *A bioinformatics approach for identification of the core ontologies and signature genes of pulmonary disease and associated disease*. *Gene Reports*, 24, 101206.
8. **H. Rehana**, S. Asaduzzaman, R. Chakma. (2021). *3D-2D BCNN based automated feature extraction & classification for hyperspectral imaging*. In *ICIEV & icIVPR*.
9. M.R. Ahmed, **H. Rehana**, S. Asaduzzaman. (2021). *Protein interaction network and drug design of stomach cancer and associated disease: a bioinformatics approach*. *Journal of Proteins and Proteomics*.
10. S. Asaduzzaman, M.R. Ahmed, **H. Rehana**, S. Chakraborty, M.S. Islam. (2021). *Machine learning to reveal an astute risk predictive framework for Gynecologic Cancer and its impact on women psychology: Bangladeshi perspective*. *BMC Bioinformatics*. **(IF 4.4)**
11. S. Islam, **H. Rehana**, S. Asaduzzaman, S.M. Hossen, R. Hossain, T. Bhuiyan, M.S. Uddin, N. Akter. (2020). *Automated risk prediction by measuring pneumothorax size using deep learning*. In *2020 IEEE Region 10 Symposium (TENSYP)* (pp. 1747–1751). IEEE.
12. **H. Rehana**. (2017). *Bangla handwritten digit classification and recognition using SVM algorithm with HOG features*. *Proceedings of the IEEE Conference on Electrical and Computer Engineering (EICT)*.

PREPRINTS & MANUSCRIPTS IN PREPARATION

- **H. Rehana**, Y. He, J. Hur. (2026). *Evidence-Grounded Evaluation of Fine-Tuned Large Language Models for Vaccine Adjuvant Extraction from Infectious Disease Literature*. (Manuscript in preparation).
- **H. Rehana**, J. Zheng, Y. He, J. Hur. *CO-PICO/PECO: An Occupational Condition-based PICO/PECO Framework for Structuring Occupational Safety and Health Evidence using Generative Pretrained Transformers*. (Manuscript in preparation).
- N.B. Çam*, **H. Rehana***, J. Zheng, B. Bansal, Y. He, J. Hur, A. Özgür. (2024). *Ontology-based protein–protein interaction explanation using large language models*. bioRxiv.
- **H. Rehana**, M. Ibrahim, H. Ali. (2023). *Plant disease detection using region-based CNN*. arXiv.

ORAL PRESENTATIONS

- **H. Rehana**, Y. He, J. Hur. *Schema-Grounded Multitask Instruction Fine-tuning for Joint Biomedical Named Entity Recognition and Relation Extraction*. International Workshop on Vaccine and Drug Ontology Studies (VDOS), International Conference on Biological and Biomedical Ontology (ICBO), 2026.

- **H. Rehana**, J. Zheng, Y. He, J. Hur. *Condition-based occupational PICO/PECO and its applications for LLM extraction and ontological modeling of structured occupational safety and health*. International Conference on Intelligent Biology and Medicine (ICIBM), 2026. (Flash talk; Accepted with Travel Award)
- **H. Rehana**, J. Zheng, F.Y. Yeh, B. Bansal, N.B. Çam, C. Jemiyo, B. McGregor, A. Özgür, Y. He, J. Hur. *Leveraging large language models for cancer vaccine adjuvant name extraction from biomedical literature*. American Medical Informatics Association (AMIA) Annual Symposium 2025. (**Travel Award**)
- **H. Rehana**, J. Zheng, J. Hur and Y. He. *PICO/PECO-Guided Large Language Model Extraction and Ontological Modeling of Structured Occupational Safety and Health Evidence from Biomedical Literature*. 16th International Conference on Biological and Biomedical Ontology, Nov 9, 2025. (Virtual)
- **H. Rehana**, J. Zheng, L. Yeh, B. Oğuzoğlu, A. Özgür, Y. He, and J. Hur. *Automated Extraction and Classification of Vaccine Adjuvant Mechanisms Using Large Language Models*. 14th International Workshop on Vaccine and Drug Ontology Studies (VDOS), Nov 7, 2025. (Virtual)
- **H. Rehana**, J. Zheng, L. Yeh, B. Bansal, N.B. Çam, C. Jemiyo, B. McGregor, A. Özgür, Y. He, J. Hur. *Identifying cancer vaccine adjuvants in biomedical literature using large language models*. Great Lakes Bioinformatics Conference (GLBIO), May 12–15, 2025.
- **H. Rehana**, Y. He, J. Hur. *Cancer vaccine adjuvant name recognition from clinical trial data using large language models*. 13th International Workshop on Vaccine and Drug Ontology Studies (VDOS), July 22, 2024. University of Twente, Enschede, The Netherlands. (Virtual)
- **H. Rehana**, J. Zheng, Y. He, J. Hur. *ChatGPT to identify drug-drug interactions from biomedical texts*. 14th ICBO – VDOS Workshop, August 28–September 1, 2023. Brasília, Brazil. (Virtual)

SELECTED POSTERS

- **H. Rehana**, N.B. Çam, M. Basmaci, A. Özgür, Y. He, J. Hur. *Identifying protein–protein interaction from biomedical text using generative pre-trained transformer*. University of North Dakota Cybersecurity Awareness and Research Symposium (CARS), October 30–31, 2023. Grand Forks, ND. (**1st place; Best Poster**)
- **H. Rehana**, H. Row, J. Hur, M. Kim, C. Dyke. *A Literature-Grounded Artificial Intelligence Framework for Vascular Access Complication Risk Modeling in Transcatheter Aortic Valve Replacement*. TRANSCEND Symposium 2026. (University of North Dakota, Grand Forks; Sanford Health, Fargo.)
- **H. Rehana**, N.B. Çam, M. Basmaci, A. Özgür, Y. He, J. Hur. *Leveraging large language models for extracting protein–protein interactions from biomedical corpora*. South Dakota State University Data Science Symposium, February 5–6, 2024. Sioux Falls, SD. (**Travel Award**)

AWARDS & HONORS

- **Travel Stipend**: NIAID BRC AI Codeathon 2025 (Argonne National Lab)
- **Conference Travel Awards**: AMIA Annual Symposium 2025 (University of North Dakota) ; Data Science Symposium 2024 (South Dakota State University); International Conference on Intelligent Biology and Medicine 2026 (ICIBM)
- **Challenge Award**: BioASQ12 Task BioNNE 2024 — 2nd place in Nested Named Entity Recognition
- **Best Poster Award**: Cyber Awareness and Research Symposium 2023 (University of North Dakota)
- **ICT Master’s Research Fellowship**: ICT Division, Government of Bangladesh (2021–2022)

CERTIFICATIONS & COMPLIANCE

- **Human Subjects & Clinical Research Compliance (CITI Program)**: Active certifications (2025–2028) in Biomedical Research Investigators, HIPAA for Researchers, Good Clinical Practice (GCP – U.S. FDA Focus), Responsible Conduct of Research, and Data or Specimens Only Research.
- **Clinical Data Privacy & Governance**: Trained in HIPAA-compliant handling of sensitive patient data, clinical records, and biomedical research datasets.
- **Professional ML Training (NVIDIA Deep Learning Institute)**: Fundamentals of Deep Learning; Building Transformer-Based Natural Language Processing Applications; Applications of AI for Anomaly Detection.

SERVICE & MEMBERSHIPS

- **Reviewer**: Scientific Reports (Springer Nature) — Reviewed 2 manuscripts.
- **Memberships**: American Medical Informatics Association (AMIA), Institute of Electrical and Electronics Engineers (IEEE), Association for Computing Machinery (ACM), International Society for Computational Biology (ISCB).